

Lecture 9: Sexual Development and Behavior

REMINDERS: **Hypothalamus** controls **Endocrine (hormone)** systems via effect on adjacent **Pituitary Gland** (the “Master Gland”)

- Produces **Releasing Hormones** that flow **via blood vessels** to **Anterior Pituitary** stimulating gland to release its own hormones
- Produces other **Hormones** (e.g. **Oxytocin**) sent (like NTs) **via axons** to **Posterior Pituitary**, then circulate in bloodstream

Reproductive Hormones have **Organizing Effects** (on anatomy in fetal devel & puberty) & **Activating Effects** (influence behavior)

- **Both** sexes have female (**Estrogens**) and male (**Androgens**) hormones, just in **different proportions**
 - These Steroid Hormones are produced mainly in Ovaries/Testes, but also in Hypothalamus and Adrenal Glands
- NOTE: Typically, **males have XY** sex chromosomes, **females have XX**, but hormone activity is required to determine gender!

ORGANIZING EFFECTS

Fetal Development of Sexual Anatomy

- Every mammalian fetus has the **anatomical precursors** for BOTH sexes
 - e.g. **Gonads**, earliest sex structure to develop, same initial structure will **become Testes or Ovaries**
 - e.g. **Genitalia**, same initial structures in all fetuses, develop into **male or female** anatomy, depending on Androgens/Not
 - e.g. Each fetus has both **Wolffian and Mullerian** ducts; Their development depends on Androgens/Not
 - These become either **Vas Deferens & Prostate** or **Fallopian Tubes & Uterus** respectively; Other system degenerates
- The **genes** controlling male/female body & brain development are also present in BOTH sexes, EXCEPT...
 - The “**switch**” is on the male’s **Y Chromosome**; it signals production of the **Testis-Determining Factor (TDF)** Enzyme
 - Occurs during **Critical Period** of fetal development (TDF appears : 6th–8th week; Genitals developed by 4th month)
- **If TDF is NOT present => Female**
 - Ovaries differentiate, Mullerian system develops, Wolffian regresses, female genitalia develop – **regardless of genotype!**
 - If fetus is **XY**, but lacks specific gene for TDF (or other TDF deficit), will develop internally & externally as **female**
 - If fetus is **XO (Turner’s Syndrome)**, no Y Chromosome) will develop internally & externally as **female**
 - Both **non-XX** above will be **infertile**, however, since two Xs required to produce ova (eggs)
- **If TDF is present => Male**
 - **Testes** differentiate, producing **Androgens**, including **Testosterone** => Wolffian ducts and male genitalia develop
 - Testes also produce **Anti-Muellerian Hormone**, inhibits development of Mullerian system
 - If **XY** fetus is **Androgen-Insensitive**, it’s testes still produce Androgens and Anti-Muellerian Hormone, but since its Wolffian system’s **androgen receptors are absent**, it not develop, **and** its Mullerian System is inhibited
 - So individual has no internal sex organs (& so is infertile) except rudimentary, internal testes
 - But external **body develops as a female**, although without androgen-stimulated pubic, or other secondary, hair
 - If **XX** fetus is exposed to **Testosterone** during critical period, develops male, or semi-male form, sometimes infertile
 - Some tendency for these individuals to be homosexual, tho per body, brain, and/or societal influences??
- **PLUS:** **Testosterone** enters fetal cells, where it is converted (“**aromatized**”) into **Estradiol** (an **Estrogen!**) => **Male** development
 - So, why doesn’t **Mom’s estradiol** masculinize every fetus? Answer: **Apha-Fetoprotein!**
 - **Apha-Fetoprotein** in fetal/infant blood, **binds with Estrogen**, preventing it from entering fetal cells (later inactivated)
 - **Excessive Estrogens** (e.g. DES, synthetic estrogen to prevent miscarriage) can overwhelm safeguard, **masculinize fetus**

Secondary Sexual Characteristics at Adolescence

- **In both sexes**, **Hypothalamus** releases **Gonadotropin-Releasing Hormones (GnRH)**, causing **Anterior Pituitary** to release the **Gonadotropic Hormones: Lutenizing Hormone (LH)** and **Follicle Stimulating Hormone (FSH)**
- **In Males**, these hormones >> Testes produce **sperm** and **Testosterone** (and other Androgens, and low levels of Estrogens)
 - >> facial & other secondary hair growth (& later baldness), muscular development, enlargement of larynx, stop bone growth
- **In Females**, these hormones >> Ovaries produce **ova** and **Estradiol** (and other Estrogens, and low levels of Androgens)
 - >> breast development, alteration of fat deposits, menstrual cycle of egg release & uterine build-up/decline, stop bone growth
 - Female **secondary hair growth** via **Androstenedione** (an **Androgen**) released by **Adrenal Glands**

Sexual Differences in Brain Development

- Presence/Absence of **Testosterone** during prenatal period and early infancy => differences in brain
 - **Medial Preoptic Area (MPOA)** of Hypothalamus, has Androgen receptor sites, is esp active during **Male sexual behavior**
 - This area includes the **Sexually Dimorphic Nucleus (SDN)** which is **2.5X larger in Males** than Females
 - **Early Testosterone** is **required** for this devel, and w/o it adult Male will not respond normally to androgen activity
 - **Ventro-Medial Hypothalamus (VMH)**, has Estrogen receptor sites, is especially active during **Female sexual behavior**
 - Develops (esp sensitivity to estrogens) in **absence of early Testosterone** (Area also implicated in **control of eating**)

- Other sexual dimorphisms in brain anatomy (may be related to early Testosterone levels, but insufficient data) include:
 - **Connectivity** patterns in cortex, per diffusion-based imaging
 - Males show more intra(within)-hemispheric connections while females show more inter(between)-hemispheric
 - Suggests males may better integrate perception & action, females better integrate analytic & intuitive processing
 - May help account (with Culture) for better spatial abilities in males, better communication abilities in females
 - **INAH3**, part of Sexually-Dimorphic Nucleus, larger in Heterosexual Males, smaller in Females and Homosexual Males
 - In homosexual males, not clear if such differences are cause or effect of behavior?

ACTIVATING EFFECTS

Neural and Hormonal Activation of Sexual Behavior - All depend on pre-established “Organizing Effects”

- *In Males:*

- **Medial Preoptic Area (MPOA)** of Hypothalamus, including Sexually Dimorphic Nucleus, is critical for sexual behavior
 - Releases **GnRH** (Gonadotrophin Releasing Hormone) > Anterior Pituitary releases the Gonadotrophins LH & FSH
 - These Gonadotrophins travel through bloodstream > Testes release Testosterone
 - Testosterone feeds back to MPOA, escalating arousal
 - Circuit includes **VTA** (Ventral Tegmental Area) > **Nucleus Accumbens** (*Pleasure!*) near Basal Forebrain
 - Releases **Dopamine** to Nucleus Accumbens in response to sexual stimulation
 - Area also implicated in addiction to amphetamines, cocaine, and chocolate; Rats will stimulate area to death
 - MPOA also stimulates Basal Ganglia which communicates with Spinal Nucleus of the Bulbocavernosus (SBN)
 - Motor neurons of SBN => rhythmic contractions for ejaculation
 - At orgasm, MPOA signals Posterior Pituitary to release **Oxytocin**
 - After ejaculation, Anterior Pituitary releases **Prolactin**, producing Refractory Period before male can respond again
- NOTE: Hormone profile in human males shows high levels of “bonding hormones” like Prolactin, Oxytocin, Vasopressin
 - Similar profile to that seen in certain other mammal species in which males contribute to child care.
- **MPOA** also responds to input from **Medial Amygdala**, implicated in **Aggression** (see Emotion lecture)
 - Some correlation between high Testosterone and Aggression (as in violent crime, tho not nec sexual crime)
- NOTE: Medial Amygdala also likely receives Pheromone input, as well as Somatosensory info from genitals
 - In rodents, smell can inhibit aggression toward females, increase aggression toward rival males
- Also responds to input from Cerebral Cortex (**Learning** plays a greater role in human sexual activity than in nonhumans)
 - Includes Sensory (including visual identification), Memory, Prefrontal evaluation, and Motor organization
 - Can lead to idiosyncratic partner-, place- or object-specific sexual responses

- *In Females:*

- **Androstenedione** (chemically like Testosterone), an **Androgen** produced by adrenals, for sexual motivation
 - Gets converted into Testosterone in bloodstream, activates **MPOA**
- Then, just as in Males, **MPOA** > **GnRH** > **LH & FSH** > stimulates Ovaries and Adrenals (short-term positive feedback)
 - Estrogens from Ovaries stimulate **Ventro-Medial Hypothalamus (VMH)**,
 - This is region most activated during female sexual behavior
- VMH and MPOA stimulate pleasure circuit: **VTA** releases Dopamine to **Nucleus Accumbens** for reinforcement
 - > Basal Ganglia > SBN for rhythmic contractions, as in males
- VMH also stimulates **Periaqueductal Gray Area** which produces Endorphins, in part to suppress Pain
 - Also signals Posterior Pituitary to release **Oxytocin** (at time of orgasm)
 - Oxytocin also linked to release of milk in lactating females, aids in bonding with offspring
- After sex, females do not show same Prolactin release or Refractory Period
 - Note: Instead, **Prolactin** in pregnant females, stimulates milk production
- **VMH** also responds to input from Medial Amygdala (Pheromones) and from Cerebral Cortex (**Learning** etc)

Role of Pheromones in Mediating Sexual Behavior

- **Pheromones** = Hormones released by one individual that affect behavior/physiology of conspecific; Found in sweat of humans
 - In most mammals, detected by **Vomerinal Organ (VNO)** – specialized olfactory receptors, respond only to pheromones
 - Direct connections to Medial Amygdala and Medial Preoptic Area of Hypothalamus
 - VNO controversial in humans: Exist / Not? Vestigial / Functional?
- However, humans DO appear to respond to pheromones (Pheromones *control* rat behavior, *influence* human behavior)
 - In Women: Sweat in alcohol swabbed on upper lip of other women (!) vs. placebo => synchronized menstrual cycles
 - In Men: Aftershave spiked with male pheromone vs. placebo => increased # of sexual interactions (inc'd attractiveness)